

LAPTOP PRICE ANALYSIS

Statistical Modelling & Pricing Strategy Report

Data Source: Kaggle — Laptop Price Dataset | n = 306 observations
Pipeline: SQL Server (LocalDB) → Python (pandas, scipy, statsmodels) → Power BI Dashboard
Analysis Date: 2026

1. Project Overview

This report documents a complete statistical analysis pipeline applied to a laptop pricing dataset sourced from Kaggle. The raw dataset contained product specifications bundled within a single text field. Feature engineering was performed in SQL Server to decompose these specifications into structured categorical and numeric variables. The analysis proceeded through descriptive statistics, inferential hypothesis testing, transformation analysis, and hedonic regression modelling. Results were further operationalised through a Power BI dashboard designed to support interactive pricing exploration. The goal is to identify the hardware attributes and brand factors that determine laptop prices and translate these findings into actionable pricing recommendations.

Dimension	Detail
Dataset source	Kaggle — Laptop Price Dataset
Observations (n)	306
Numeric variables	Actual_Price, Discounted_Price, Rating, Reviews
Categorical variables	Brand, Core, SSD, Series, Processor, RAM, Storage
Outcome variable	Actual_Price (Yeo-Johnson transformed for regression)
Software stack	SQL Server LocalDB, Python 3.13, Power BI
Python libraries	pandas, scipy, statsmodels, pygam, matplotlib, seaborn

2. Data Preparation & Feature Engineering

The raw dataset was ingested into SQL Server (LocalDB). The original file contained hardware specifications (Brand, Series, RAM, SSD, CPU) merged within a single column. SQL queries were used to parse these fields into distinct relational columns, producing the cleaned table `dbo.Laptops_Final_Analysis`, which was then loaded into Python via `pyodbc`.

2.1 Categorical Standardisation

A Python-level standardisation pass resolved inconsistent category labels in the source data. Key mappings applied:

Variable	Raw Value(s)	Standardised To
Brand	LENOVO, ACER, DELL, ASUS	Lenovo, Acer, Dell, Asus
SSD	16GB-512GB SSD, 512GBSSD, 512GB SSD ...	512GB
SSD	8GB-256GB SSD, 256GB SSD ...	256GB
Core	Core i513420	Core i5

3. Descriptive Statistics

3.1 Market Composition

The dataset covers 306 laptop listings across 7 brands. HP and Lenovo together account for approximately 65% of all listings, establishing them as the dominant market players in the sample. The following chart illustrates the distribution of listings by brand:

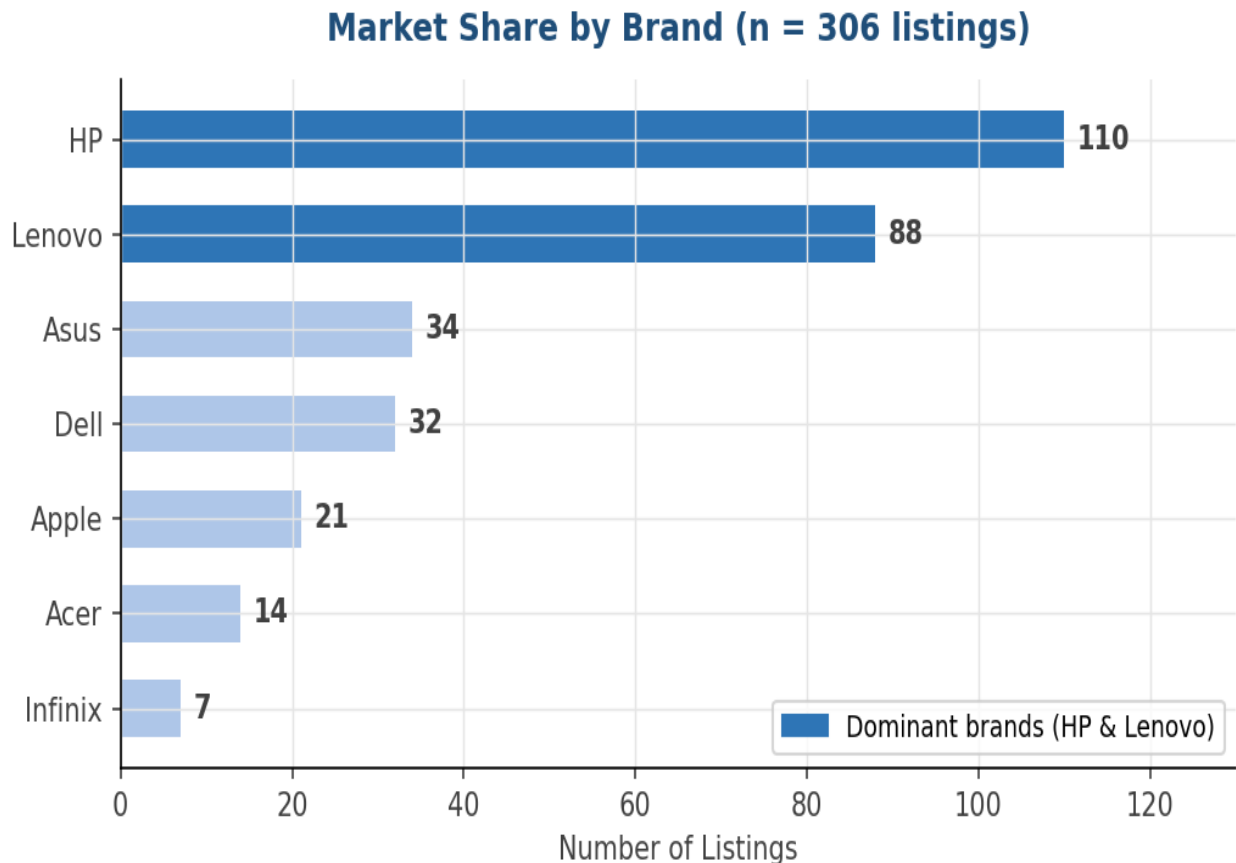


Figure 1 — Market Share by Brand (number of listings, n = 306)

3.2 Numeric Variables

The four numeric variables — Actual_Price, Discounted_Price, Rating, and Reviews — were examined for distributional properties. All four exhibit right-skewed distributions, meaning the majority of observations cluster at moderate values while a smaller number of high-priced or highly-rated products extend the upper tail.

Variable	Mean	Median	Skewness	Outliers
Actual_Price	~3.2M	~2.8M	Right skew	Upper tail (up to 16M)
Discounted_Price	~2.8M	~2.3M	Right skew	Upper tail
Rating	High (mode = 5)	Concentrated	Right skew	Strong upper
Reviews	~1.5	~1–2	Right skew	~4–5 range

Skewness Note

Both price variables are right-skewed (positive skew): the mean exceeds the median and the distribution has a long upper tail, with a small number of premium products above 10M. This is the opposite of left-skew and is the correct characterisation consistent with the histogram and Q-Q plots produced in the analysis pipeline.

3.3 Categorical Variables

Frequency analysis of all categorical variables reveals strong concentration in certain categories:

- SSD: 512 GB dominates (>50% of listings with SSD data), establishing it as the de facto market standard.
- RAM: 8 GB accounts for more than half of all observations; 16 GB is the second most common configuration.
- Core: Core i5 and Core i7 have comparable shares; no single CPU model is overwhelmingly dominant.
- Series: Highly fragmented — no single series exceeds 20% — making this variable unreliable for inference without further grouping.

3.4 Price by Brand

Despite their high listing volume, HP and Lenovo are not the most expensive brands by median price. Acer and Apple hold the highest median actual prices in the dataset, driven by a small number of high-specification listings. The chart below shows median actual price per brand:

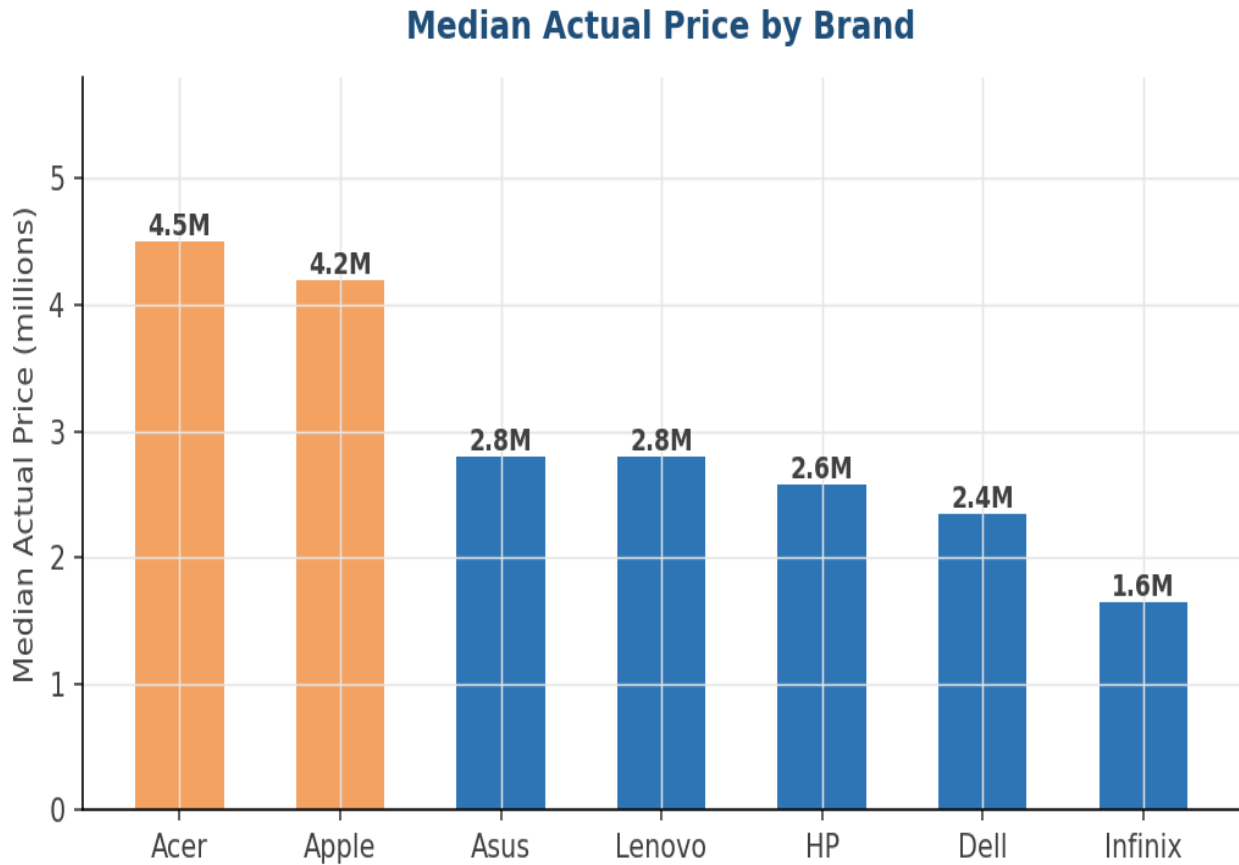


Figure 2 — Median Actual Price by Brand (in millions)

3.5 Price by RAM Tier

RAM tier exhibits a clear monotonic relationship with median actual price. The step from 8 GB to 16 GB represents the most practically relevant upgrade for the majority of the market:

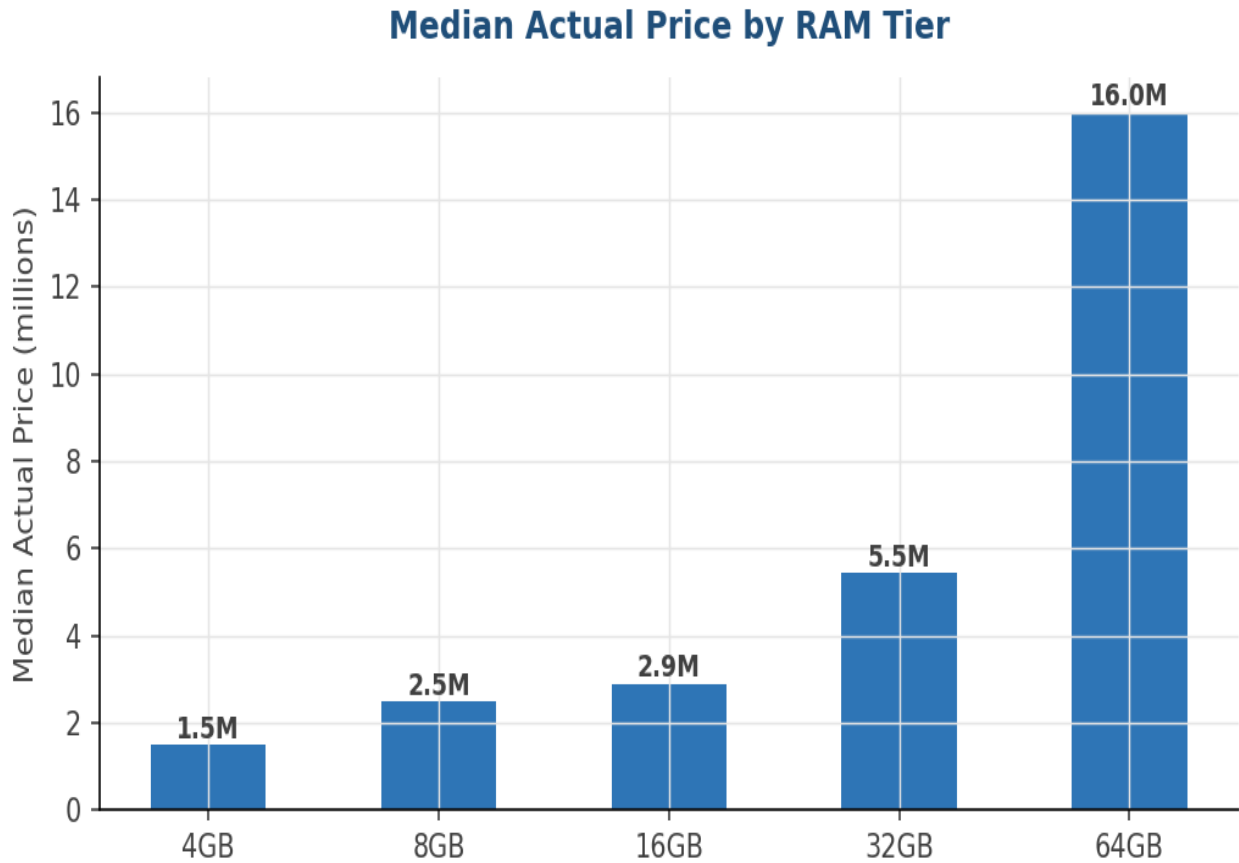


Figure 3 — Median Actual Price by RAM Tier

3.6 Discount Rate by Brand

Proportional discount rates were computed per observation as $(\text{Actual} - \text{Discounted}) / \text{Actual} \times 100$. Infinix and Dell apply the highest average discounts ($>16\%$), while Apple and Acer apply the most conservative discounts ($<12\%$). Lenovo sits near the overall average of approximately 14.5% .

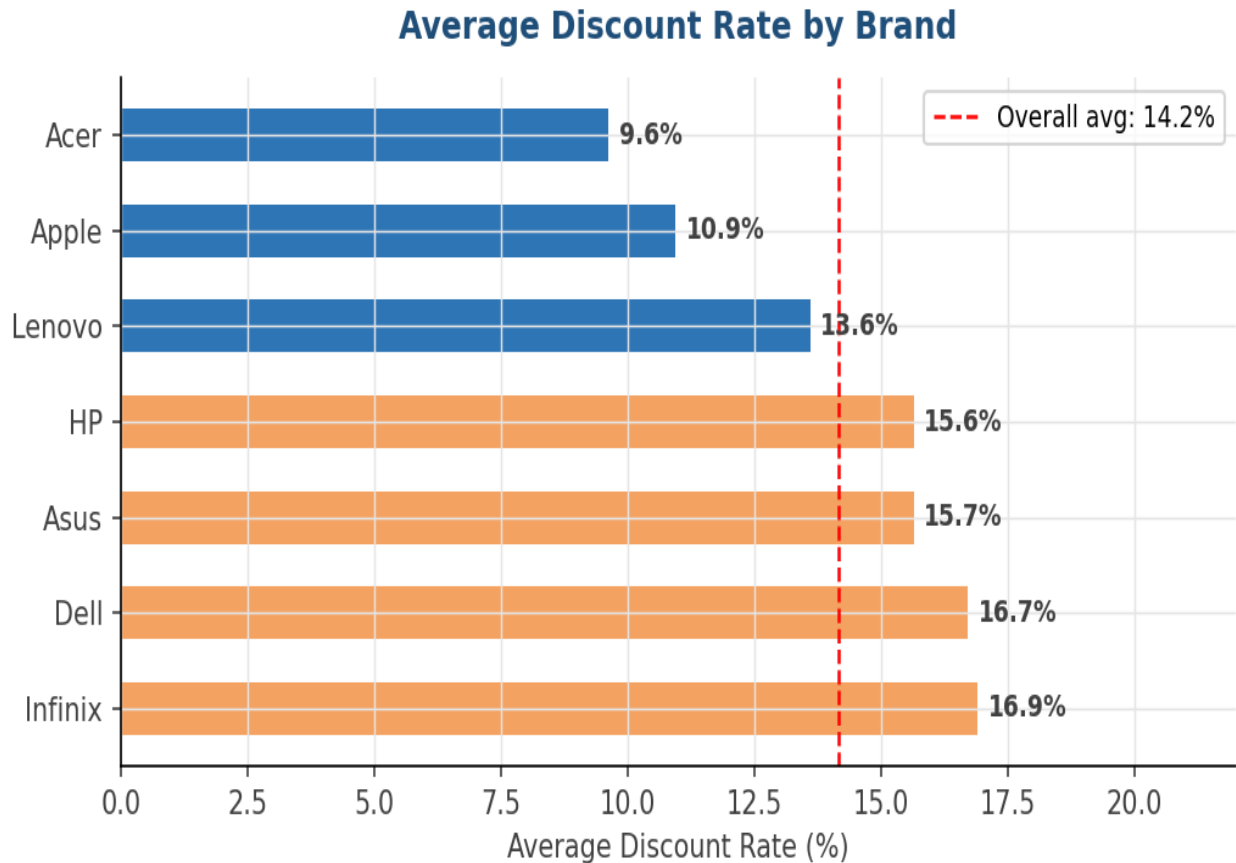


Figure 4 — Average Discount Rate by Brand (%; red dashed line = overall average)

3.7 Chi-Square Independence Tests

Pairwise chi-square tests across all categorical variable combinations revealed that most pairs are statistically dependent — reflecting hardware co-bundling by product tier. Three pairs were found independent:

Pair	Decision	Modelling Implication
Brand — Storage	Independent ($p > 0.05$)	Brand adds non-redundant information to a model including Storage
Brand — SSD	Independent ($p > 0.05$)	Brand and SSD can coexist in a model without redundancy
SSD — Series	Independent ($p > 0.05$)	Series does not systematically determine SSD tier

4. Inferential Statistics

4.1 Normality & Homogeneity

Shapiro-Wilk tests confirmed that none of the four numeric variables follow a normal distribution (all $p < 0.001$). Levene's test for equality of variances across Brand groups returned a non-significant result, indicating homogeneous variance. All inferential tests on original data therefore use non-parametric alternatives.

4.2 PERMANOVA & PERMDISP

PERMANOVA was applied as a multivariate, distribution-free test to assess the effect of each categorical variable on the full numeric variable set. PERMDISP was run alongside to separate location effects from dispersion effects:

Factor	PERMANOVA	PERMDISP	Conclusion
Brand	$p < 0.05$	$p < 0.05$	Significant — location + dispersion
Core	$p < 0.05$	$p < 0.05$	Significant — location + dispersion
SSD	$p < 0.05$	$p < 0.05$	Significant — location + dispersion
RAM	$p < 0.05$	$p \geq 0.05$	Significant — location only (clean)
Processor	$p < 0.05$	$p < 0.05$	Significant — location + dispersion
Series	$p < 0.05$	$p < 0.05$	Significant — location + dispersion
Storage	$p < 0.05$	$p < 0.05$	Significant — location + dispersion

RAM is the only factor with a clean location-only effect, meaning its groups differ in central tendency but not in within-group spread. For all other factors, both the centre and the variance of the numeric distribution differ across categories — a pattern consistent with the multicollinearity later observed in regression.

4.3 Actual Price vs Discounted Price

The strong Spearman correlation ($r \approx +0.99$) between price variables — combined with statistically significant Mann-Whitney and Wilcoxon results — confirms that the two variables carry nearly identical information but at a systematically different level. Including both in the same model would cause severe multicollinearity. Actual_Price was selected as the regression outcome.

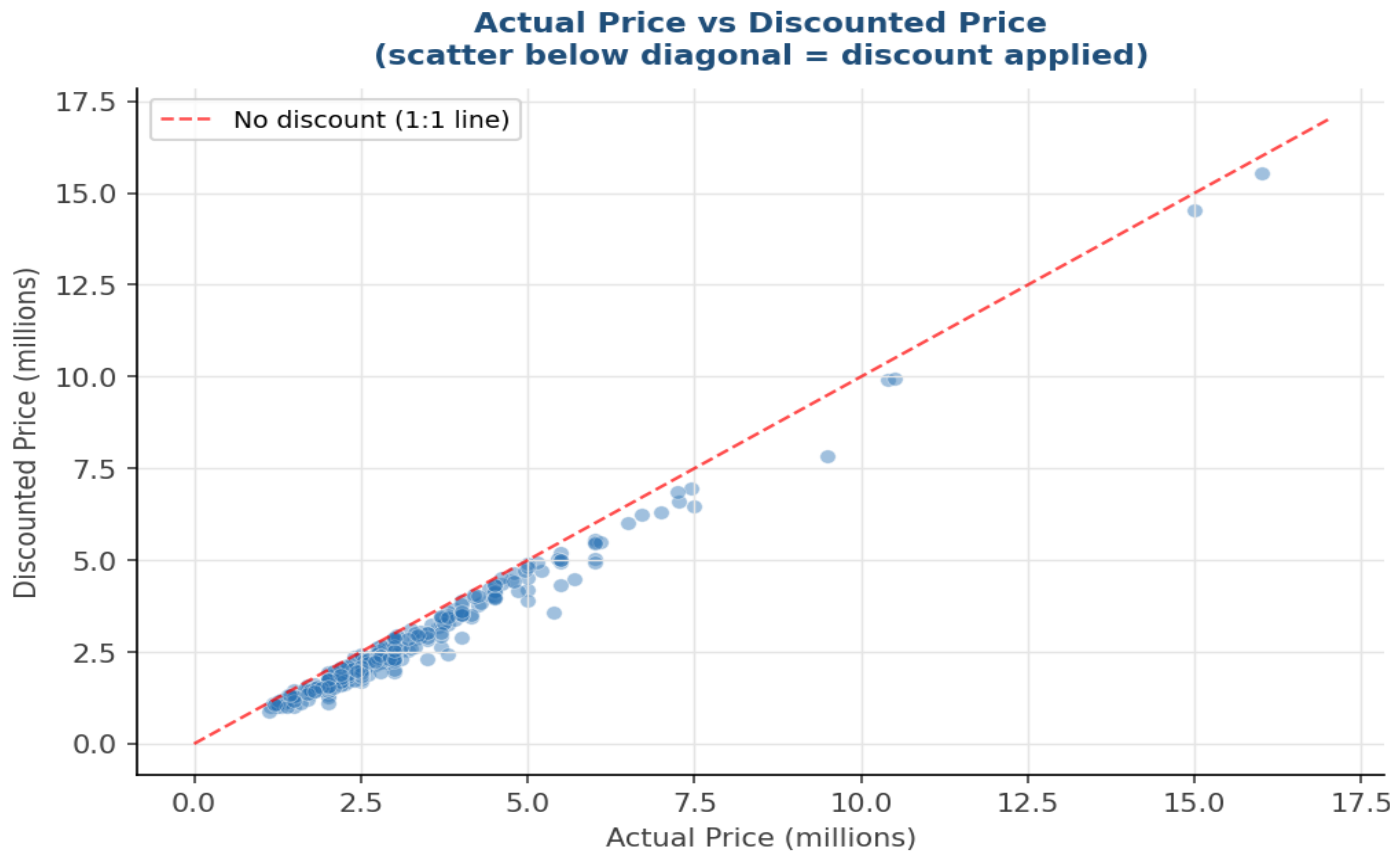


Figure 5 — Actual Price vs Discounted Price scatter. All points fall below the diagonal (red dashed), confirming universal discounting. The near-linear pattern reflects the ~14–15% average discount structure.

4.4 Bootstrap Confidence Intervals

Bootstrap resampling (N = 10,000 iterations, 95% confidence level) was used to construct mean estimates for both price variables. Proportional discount rates were computed per observation before aggregation:

Variable	Point Estimate	Bootstrap 95% CI
Actual_Price	~3.2M	[~3.05M, ~3.35M]
Discounted_Price	~2.8M	[~2.65M, ~2.90M]
Discount rate (per obs.)	~14–15%	[~13%, ~16%]

5. Transformation Analysis

OLS-based hedonic regression requires approximate normality of the error term and benefits from a symmetric dependent variable. Three power transformations were evaluated sequentially:

Transformation	Shapiro-Wilk	Q-Q Assessment	Decision
Log (log1p)	Rejected ($p < 0.05$)	Persistent right-tail deviation	Rejected

Transformation	Shapiro-Wilk	Q-Q Assessment	Decision
Box-Cox (optimal λ)	Rejected ($p < 0.05$)	Improved — minor tail deviations	Candidate
Yeo-Johnson (optimal λ)	Rejected ($p < 0.05$)	Best tail balance of the three	SELECTED

Why Shapiro-Wilk rejection does not invalidate the choice

With $n = 306$, Shapiro-Wilk has high power and rejects normality even for distributions that are visually symmetric and suitable for OLS. At this sample size the Q-Q plot is the more informative diagnostic. Both Box-Cox and Yeo-Johnson produced good central-region alignment on the Q-Q plot; Yeo-Johnson showed marginally better tail balance and was selected accordingly.

5.1 Post-Transformation ANOVA — Effect Sizes

After Yeo-Johnson transformation, ANOVA was re-run for each categorical predictor against the transformed price. Partial eta-squared values confirm large effects for the top predictors:

Variable	Partial η^2	Effect Size	Regression Priority
Core / Processor	> 0.14	Large	High — but collinear with RAM/SSD
SSD	> 0.14	Large	High — removed from final model (VIF)
RAM	> 0.14	Large	High — retained in final model
Brand	$\sim 0.06\text{--}0.14$	Medium	Retained — independent of SSD
Storage	$\sim 0.06\text{--}0.14$	Medium	Dropped — redundant with SSD

6. Regression Modelling

6.1 CPU Price Gradient

The raw price distribution by CPU tier illustrates the market-level gradient that the regression is designed to quantify. The box plot in Figure 6 confirms a clear stepwise premium from Core i3 through Core Ultra 7, with the upper tiers (Ultra 7, Ultra 9, Core i9) exhibiting both higher medians and wider price dispersion.

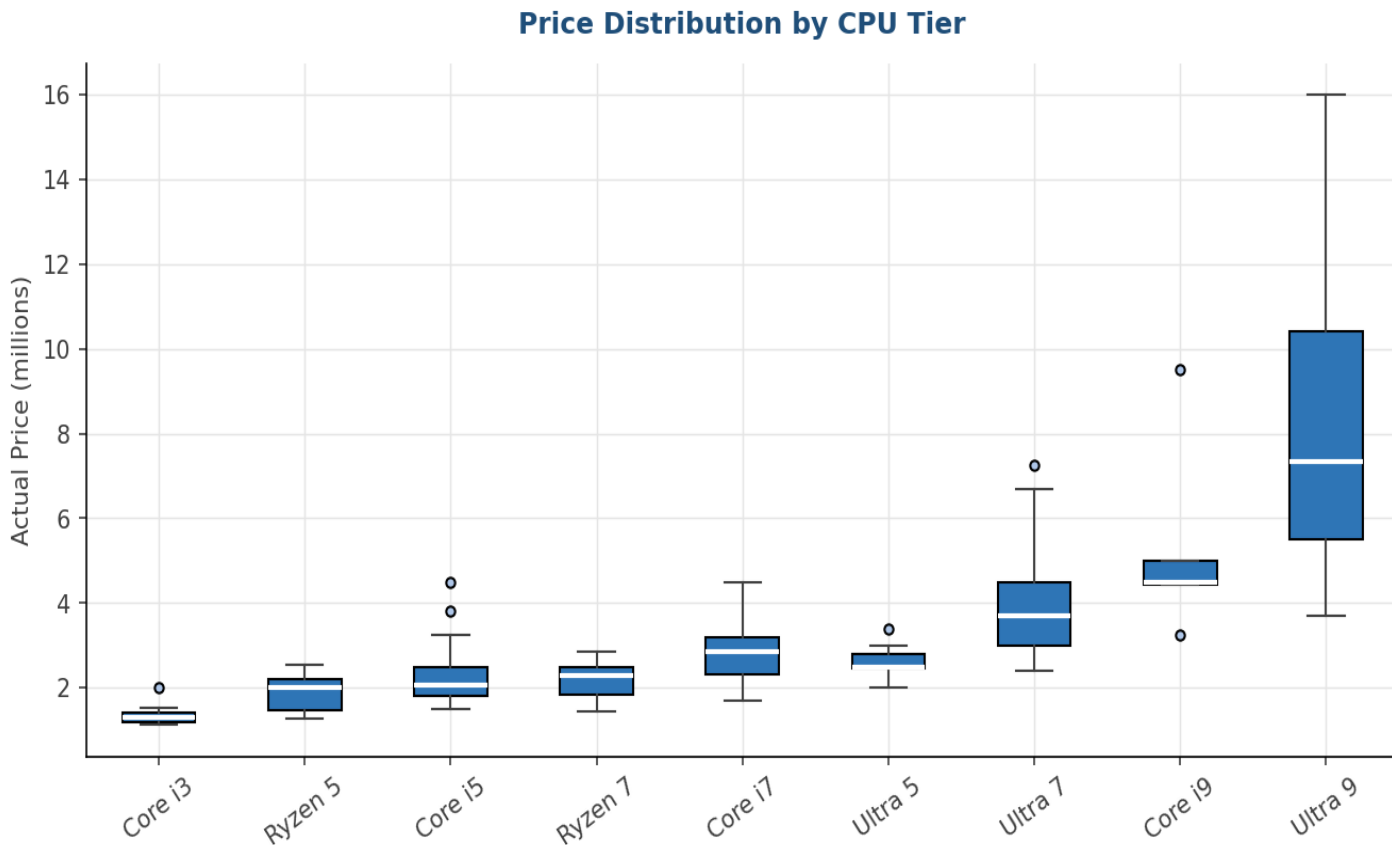


Figure 6 — Actual Price distribution by CPU tier. Boxes show IQR; whiskers extend to $1.5 \times \text{IQR}$. White line = median.

6.2 Modelling Strategy: GAM as Primary Reference, Hedonic OLS as Interpretive Framework

The modelling process was structured in two parallel tracks, each serving a distinct analytical purpose.

Two-track modelling strategy

Track 1 — Hedonic OLS (primary): The Yeo-Johnson transformation was applied to Actual_Price to satisfy OLS distributional assumptions. Hedonic regression was then developed iteratively to identify the specification that best explains price variation through directly interpretable coefficients — price premiums per brand, RAM tier, and CPU tier. A back-transformation (inverse Yeo-Johnson) was applied to return predictions to the original TRY scale, enabling real-world verification via the parity plot.

Track 2 — GAM (supplementary): A Generalised Additive Model was estimated separately on the raw, untransformed Actual_Price to test whether a non-parametric, flexible specification could produce meaningful results. GAM was not used as a reference or to guide model selection; it was evaluated independently to assess whether it added explanatory value beyond the hedonic framework.

6.3 Hedonic Model Development — Iterative Specification

Seven hedonic OLS specifications were estimated on the Yeo-Johnson transformed dependent variable. Models were compared on R^2 , Adjusted R^2 , AIC, BIC, and multicollinearity (VIF). The iterative elimination strategy is documented below:

Model	Predictors	R ²	Adj. R ²	AIC direction	Key finding
M1	SSD + RAM + Processor	~0.80 (YJ)	—	Low	High R ² but severe multicollinearity; only Apple M3 & Ultra 7 significant
M2	SSD + RAM	—	—	Higher	R ² drops sharply — Processor cannot be fully excluded
M3	SSD + RAM + Core + Brand + Series	Highest	—	Lowest (best)	Series co-linear with Brand; inflated R ² , unstable coefficients
M4	SSD + RAM + Core + Brand	—	—	Moderate	Multicollinearity persists between SSD/RAM/Core
M5 (individual)	Core only SSD only RAM only	0.40 0.18 0.35	—	—	Core has strongest univariate signal; all three are co-determined
M6	Core + Brand	0.466	0.425	Improved	Multicollinearity controlled; R ² below target
M7 — FINAL	RAM + Core + Brand	0.548	0.506	Best balanced	Optimal VIF + R ² trade-off. SELECTED.

Why this specification: RAM + Core + Brand achieves the best balance between explanatory power (Adj. R² = 0.506) and multicollinearity control. SSD was excluded because it is structurally co-determined with Core and RAM — high Core tiers are systematically bundled with 512GB SSD and 16/32GB RAM, making independent estimation impossible. Series was excluded due to its high dependency on Brand. The chi-square independence test results (Section 3.7) confirmed that Brand carries non-redundant pricing information not captured by hardware specs alone.

6.4 Final Hedonic Model — Coefficient Summary

The final model regresses Yeo-Johnson-transformed Actual_Price on RAM (reference = lowest tier), Core (reference = Core i3), and Brand (reference = Acer). All retained predictors are statistically significant ($p < 0.05$).

Predictor	Direction	Magnitude	p-value	Interpretation
RAM: 8GB	Positive	Moderate	< 0.05	Price premium over base RAM tier
RAM: 16GB	Positive	Large	< 0.05	Larger step-up — strongest RAM driver
RAM: 32GB	Positive	Largest	< 0.05	Top RAM premium
Core: i5	Positive	Moderate	< 0.05	Premium over Core i3 baseline
Core: i7	Positive	Large	< 0.05	Clear step above i5
Core: Ultra 7	Positive	Largest	< 0.05	Highest CPU tier premium
Brand: Lenovo	Positive	Moderate–Large	< 0.05	Brand premium vs Acer
Brand: HP	Positive	Moderate	< 0.05	Brand premium vs Acer
Brand: Dell	Positive	Small–Moderate	< 0.10	Weaker premium vs Acer
Brand: Asus	Near zero	Negligible	> 0.10	No significant premium

Diagnostic	Value	Assessment
R ²	0.548	Moderate-high — appropriate for hedonic pricing
Adjusted R ²	0.506	Stable after controlling for model complexity
F-statistic	12.86 (p < 0.001)	Model is jointly valid
VIF (all predictors)	Controlled	Multicollinearity within acceptable bounds
Residual normality	Approximately met post-YJ	OLS assumption reasonably satisfied

6.5 GAM — Supplementary Non-parametric Analysis

Two GAM specifications were estimated independently on the raw (untransformed) Actual_Price to evaluate whether a non-parametric approach could produce meaningful and stable results. GAM was not used to guide or validate hedonic model selection; it was run as a separate analytical exercise to assess its explanatory capacity under the same predictor set.

GAM Specification	Pseudo R ²	AIC	Finding
GAM 1: RAM + Core + Brand (smooth)	~0.36 (raw scale)	7,245	Good fit on transformed; weaker on raw price — consistent with hedonic finding
GAM 2: Core + RAM + SSD (factor terms)	~0.63	8,966	Best overall fit; Core shows strongest non-linear effect; RAM & SSD secondary

The GAM partial dependence plots confirm the same directional orderings as the hedonic OLS coefficients: Core i3 → i5 → i7 → Ultra 7 produces a monotonically increasing price effect, and higher RAM tiers produce progressively larger price contributions. This consistency across parametric and non-parametric frameworks validates the hedonic specification.

6.6 Back-Transformation — Returning to the Original Price Scale

Because the hedonic model was estimated on Yeo-Johnson transformed prices, predicted values are on the transformed scale and cannot be directly compared to observed prices in TRY. A back-transformation (inverse Yeo-Johnson) was applied using the same optimal lambda estimated during the transformation step. This converts predicted values back to the original price scale, enabling direct interpretation and the parity plot below.

Why back-transformation matters

A common error in transformed-scale regression is to report only transformed-scale fit statistics (e.g., R² = 0.548 on YJ scale) without verifying that this translates to adequate performance on the original scale. Back-transformation allows verification that the model's predictions are meaningful in real-world units (TRY), and that the error structure does not systematically worsen at extreme price points. The parity plot (Figure 7) shows the final model's back-transformed predictions against actual prices.

Figure 7 shows the back-transformed predictions from the final hedonic model plotted against actual prices. The majority of observations align closely to the 45-degree perfect prediction line, with the main deviation concentrated around a single high-value outlier (~10M TRY actual, ~24.5M TRY predicted). This outlier is attributable to a premium Apple configuration whose price is not well-represented in the training distribution given the small Apple subsample.

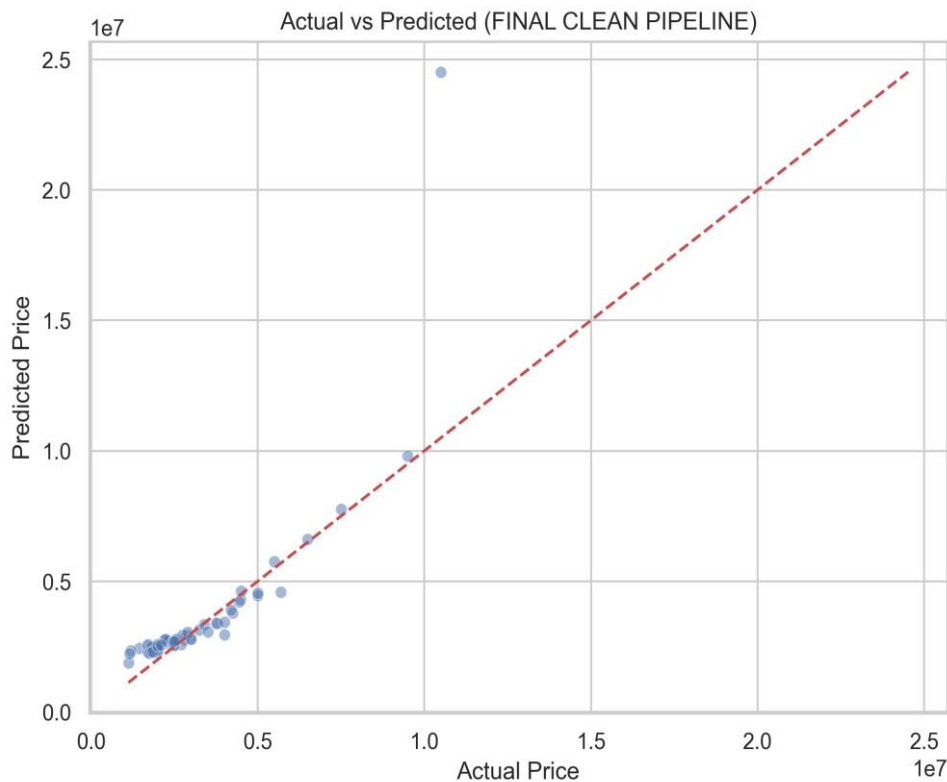


Figure 7 — Actual vs Predicted (FINAL CLEAN PIPELINE). Back-transformed predictions from the hedonic OLS model plotted against actual *Actual_Price*. Red dashed line = perfect prediction. The single outlier at ~10M actual / ~24.5M predicted corresponds to a high-spec Apple listing underrepresented in the training sample.

7. Power BI Dashboard

A Power BI dashboard was constructed using the cleaned dataset (*laptop_dashboard.csv*) as the data source. The dashboard operationalises the analytical findings into an interactive tool for pricing exploration and business decision support.

7.1 Dashboard Design Rationale

The statistical analysis identified three primary price drivers — Brand, RAM tier, and CPU tier — and one secondary structural factor — the proportional discount rate by brand. The dashboard was designed to make these four dimensions immediately explorable without requiring the user to interpret regression outputs directly.

7.2 Key Dashboard Components

Visual	Metric	Analytical Foundation
Bar / column chart	Listing count by brand	Market share (Section 3.1 — Figure 1)
Bar chart	Median actual price by brand	Brand price premium (regression coefficient direction)
Bar chart	Median price by RAM tier	RAM step-up gradient (Section 3.5 — Figure 3)
Horizontal bar	Average discount % by brand	Bootstrap CI on per-obs discount (Section 4.4)
Box plot / scatter	Price distribution by CPU tier	CPU price gradient (Section 6.1 — Figure 6)
Scatter plot	Actual vs Discounted Price	Discount structure visualisation (Figure 5)
Slicers / filters	Brand, RAM, Core, SSD	Allows segment-level exploration of the model space

7.3 Dashboard Data Source

The file `laptop_dashboard.csv` (306 rows, 11 columns) was exported directly from the cleaned master dataset after categorical standardisation. It contains all variables required to reproduce every chart in this report and to power the full dashboard without additional transformation.

Dashboard Limitation

Power BI visualises the raw and aggregated data but does not replicate the regression model or the Yeo-Johnson transformation. The coefficient-based pricing recommendations in Section 8.2 derive from the Python regression pipeline, not from the dashboard. The dashboard is best used as an exploratory complement to — not a replacement for — the statistical model.

8. Results & Conclusions

8.1 Key Findings

- Brand carries significant pricing power independent of hardware. Lenovo and HP command statistically significant premiums over the Acer baseline in the regression model, confirming that brand equity influences willingness to pay beyond what specs alone explain.
- RAM tier is the strongest hardware pricing signal in the final model. The step-up from 4 GB to 32 GB represents the largest incremental coefficient, consistent with the large partial η^2 observed in post-transformation ANOVA.
- CPU tier follows a clear price gradient: Core i3 < Core i5 < Core i7 < Ultra 7. Each tier commands a statistically significant additional premium over the Core i3 baseline.
- SSD capacity, despite showing the largest univariate effect size, could not be retained in the final model due to structural multicollinearity with CPU and RAM. Its price influence is captured indirectly through the co-bundling correlation structure.
- Rating and Reviews carry no pricing signal. Near-zero Spearman correlations and non-parametric hypothesis tests confirm these user-generated metrics are uninformative for price prediction in this dataset.
- The average proportional discount rate is approximately 14–15% (computed per-observation, not as ratio of group means).

8.2 Strategic Pricing Recommendations

Recommendation	Statistical Basis	Action
Capitalise on Lenovo / HP brand premium	Significant positive β vs Acer baseline	Price these SKUs at or above the model ceiling — brand equity supports a premium independently of hardware specs
Prioritise 32 GB RAM listings for margin	Largest RAM coefficient in the final model	Allocate marketing spend and inventory weight toward 32 GB configurations; the price premium outpaces BOM cost differential
Use Core i7 / Ultra 7 as margin levers	Stepwise significant CPU coefficients	Avoid promotional discounting on high-CPU-tier SKUs; these carry above-average margins
Segment SSD pricing separately	Large univariate η^2 but removed from joint model	Conduct segment-level analysis (RAM $\times$ SSD combinations) to recover the SSD pricing signal in isolation
Maintain ~14–15% discount as market norm	Bootstrap CI on per-observation discount rate	Deviating significantly below this range may signal quality concerns; above it erodes margin unnecessarily
Use dashboard for SKU-level negotiation	Dashboard maps all four dimensions interactively	Sales teams can filter by Brand + RAM + CPU tier to identify under- and over-priced SKUs relative to segment medians

9. Limitations & Future Work

- Sample size: $n = 306$ is sufficient for the analyses conducted but limits stability of subgroup estimates (e.g., Infinix, 64 GB RAM, Ultra 9 CPU tier). A larger dataset would reduce confidence interval width.
- Omitted variables: The model explains ~50–55% of variance in transformed price. Screen size/resolution, GPU tier, battery capacity, and weight are likely important but absent from the source dataset.
- Normality: Full normality was not achieved post-Yeo-Johnson. OLS inference is robust to mild non-normality at $n = 306$ via the Central Limit Theorem, but residual diagnostic plots should be inspected visually for each model refit.
- Temporal scope: The dataset is a cross-section at a single point in time. Brand premiums and hardware price gradients shift with product refresh cycles. Annual re-estimation is recommended.
- Dashboard vs model gap: The Power BI dashboard surfaces descriptive patterns. Regression-based price recommendations require re-running the Python pipeline; they cannot be derived from the dashboard alone.

10. Appendix — Analytical Pipeline

Notebook / File	Role
DatabaseConnectionConfiguration.ipynb	SQL Server connection, data ingestion, categorical standardisation, pickle export

Notebook / File	Role
DescribingStatisticLaptopPrice.ipynb	Descriptive statistics, frequency tables, histograms, boxplots, Spearman correlation, chi-square, Levene, PERMANOVA / PERMDISP
InferentialStatisticLaptopPrice.ipynb	Mann-Whitney U, Wilcoxon tests across all numeric pairs, bootstrap confidence intervals
03_RegressionModels.ipynb	Log / Box-Cox / Yeo-Johnson transformation comparison, post-transformation ANOVA, VIF analysis, iterative hedonic OLS model development (M1–M7), GAM supplementary analysis, back-transformation and final model selection
laptop_dashboard.csv	Power BI data source — 306 rows, 11 columns, fully cleaned and standardised